

## Problem Set 1

---

All answers should be brief.

### I. SPECULATION

In 2015, the journal *Science* published an article by two Johns Hopkins medicine professors, Cristian Tomasetti and Bert Vogelstein, called “Variation in Cancer Risk among Tissues Can Be Explained by the Number to Stem Cell Divisions.” In the abstract, the authors say

These results suggest that only a third of the variation in cancer risk among tissues is attributable to environmental factors or inherited predispositions. The majority is due to “bad luck.”

1. Suppose that it is the day after the article was published (January 3, 2015) and that you believe the market is semi-strong-form efficient. You believe that the paper implies that cigarette smoking is unlikely to cause cancer (bad luck causes cancer) and that, as a result, more people will take up smoking in the future.<sup>1</sup> Can you speculate on this view in securities markets? If so, how?
2. Now suppose that you believe that the market will over-react to the publication of the paper (so it is not semi-strong form efficient). In particular, you think that the prices of stocks in cigarette company stocks will rise sharply in the short-term (days), but will fall back still below current levels in the medium horizon (months).

---

<sup>1</sup>What is wrong with this logic?

- a) How would you speculate on this view using cigarette stocks?
- b) What are some challenges to implementing your strategy?
- c) What properties could a derivative instrument have to address those challenges?

Hint: Look up an exotic option called a “lookback option with floating strike.”

3. On January 15, 2015, the Swiss National Bank (Switzerland’s central bank) announced that it would abandon its currency peg of 1.20SFr./€. (See, e.g., the *FT* story “Swiss Stun Markets and Scrap Franc Ceiling” for details.) By all accounts, the change was unanticipated.

- a) Suppose it is January 14 and that you have inside information about the SNB’s decision. You believe that this will cause the Swiss Franc to appreciate against the Euro. Carefully describe the position you would take to speculate on this view.
- b) Again, suppose it is January 14 and you have inside information about the SNB’s decision. Now, though, you believe that the announcement will increase volatility in the Franc-Euro exchange rate. (You do not have a view on the direction that the rate will move.)

Do you think that you can take a position in forwards to speculate on this view? Why or why not?

## II. ARBITRAGE

For each of the following examples, say whether there is an arbitrage. If there is an arbitrage, show it formally. If there is not (or not enough information to say that there is), explain why not.

1. There is a security that costs \$1 today and pays off \$10 for sure in a year.
2. There is a security that costs \$10 today and pays off \$1 for sure in a year.
3. As above, there is a security that costs \$1 today and pays off \$10 for sure in a year. There is also a bond with yield 10%.
4. There is a security that costs \$1 today and pays off \$10 on average in a year. There is also a bond with yield 10%.

### III. THE LAW OF ONE PRICE

Consider a derivative,  $X_t$ , written on gold that gives the right to redeem for one troy ounce the first time,  $\tau$ , that the price,  $p_t$ , of a troy ounce reaches \$3000.

Assume (i) that the price will reach \$3000 eventually and (ii) that there are no storage costs or transaction costs of any kind.

1. Go here: <https://www.cmegroup.com/markets/metals/precious/gold.html>.  
What is the price of a troy ounce of gold today?
2. What is the price,  $X_0$ , of the derivative described above?
3. Show explicitly that if  $X_0$  is different from your answer above, there is an arbitrage opportunity.
4. How do you think your answer could change if gold were costly to store? Explain briefly.

### IV. CONVEXITY (MATH PRACTICE)

Let  $\mathbf{X}$  be a random variable, for example a stock price. Let  $\mathbb{E}[\mathbf{X}]$  denote its expected value and  $\text{Var}[\mathbf{X}]$  its variance.

1. Show that  $\mathbb{E}[\mathbf{X}^2] \geq \mathbb{E}[\mathbf{X}]^2$ .

Hint: Use the definition of the variance  $\text{Var}[\mathbf{X}] = \mathbb{E}[(\mathbf{X} - \mathbb{E}[\mathbf{X}])^2]$  and the formula  $\text{Var}[\mathbf{X}] = \mathbb{E}[\mathbf{X}^2] - \mathbb{E}[\mathbf{X}]^2$ .

2. *Jensen's inequality* says that if a function  $f$  is convex (i.e.  $f'' \geq 0$ ) then for any random variable  $\mathbf{X}$

$$f(\mathbb{E}[\mathbf{X}]) \leq \mathbb{E}[f(\mathbf{X})].$$

Analogously, if  $f$  is concave ( $f'' \leq 0$ ), the inequality goes in the other direction.

Show by differentiation that  $x^2$  is a convex function and comment on the previous result.

3. In finance, the quantity  $L[\mathbf{X}] := \log \mathbb{E}[\mathbf{X}] - \mathbb{E}[\log \mathbf{X}]$  is sometimes called the *entropy* of a random variable  $\mathbf{X}$ . Can you say whether it is positive or negative?

Hint: Check whether  $\log$  is convex or concave and apply Jensen's inequality.